TEACH YOURSELF INTERMEDIATE XI UNIT 5

Applications & Impacts of Computing

AI, IoT, ethics, and reliable information

What you will master

- Describe IoT components and smart-city/industry examples
- Explain basics of AI and everyday uses in Pakistan
- Judge reliable vs unreliable online sources
- Discuss human and environmental impacts of computing

How to use these notes

- Read one section, then close the book and explain it aloud.
- Do every *Try it* box before moving on.
- Memorize tables and truth patterns — exams love them.

5.1 Computing in daily life

Computing spans education (LMS), healthcare (records), banking (apps/ATMs), agriculture (weather/irrigation), and industry automation.

5.2 AI & IoT

Artificial Intelligence — systems that learn patterns (recommendations, voice assistants, image filters). Needs quality data and clear purpose.

Internet of Things (IoT) — sensors + connectivity + cloud analytics (smart meters, wearables, fleet tracking).

Smart city

Traffic sensors, air quality monitors

Healthcare

Remote vitals, bed availability

Education

Automated attendance, adaptive quizzes

5.3 Digital citizenship

Evaluate sources: author, date, evidence, bias. Prefer peer-reviewed, official, and cross-checked facts over viral posts.

Impacts include e-waste, energy use, job shifts, and privacy — balance innovation with responsibility.

Teach Yourself — Practice & Check

1. Name three IoT devices at home or school and what data they collect.
2. Compare two news URLs for credibility using a checklist.